Linear Algebra
Assignment 2

Deadline: 26 October 2026

Instructions

- Submit handwritten solutions. Calculators are not allowed.
- Complete the assignment by yourself, without using AI.
- Scan your solutions and submit them as a single PDF to the Moodle course.
- Name your submission file with your roll number: `<roll-number>.pdf`.
- The deadline is **26 October 2026**. No extension of the deadline will be granted.

Reference: Lecture notes available on Moodle. Exercise numbers and references below follow the lecture notes.

Chapter 5: Vector Spaces and Subspaces

Exercise 5.16. Which of these are subspaces of $\mathbb{R}^3$? Justify each answer, and when a set fails, say which condition of Proposition 5.6 it violates.

(a) $\{(x, y, z) : x + y + z = 0\}$

(b) $\{(x, y, z) : x + y + z = 1\}$

(c) $\{(x, y, z) : xyz = 0\}$

(d) $\{(x, y, z) : x = y\}$

(e) $\{(x, y, z) : x \geq 0\}$

Exercise 5.17. Let U and W be subspaces of V. Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$. *Hint: if neither containment holds, pick $u \in U \setminus W$ and $w \in W \setminus U$ and ask where $u + w$ can live.*

Exercise 5.18. In $M_2(\mathbb{R})$, let S be the symmetric matrices and K the antisymmetric ones. Show that both are subspaces, that $S + K = M_2(\mathbb{R})$ and that $S \cap K = \{0\}$. Compare with Exercise 4.29.

Exercise 5.19. For $A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 1 & 3 & 3 \end{pmatrix}$, find generators for $N(A)$, $C(A)$ and $R(A)$, and verify your null-space generator against the original matrix.

Exercise 5.20. Show that $\{A \in M_n(\mathbb{R}) : \operatorname{trace} A = 0\}$ is a subspace. Example 5.7 showed that $\{A : \det A = 0\}$ is not a subspace of $M_2(\mathbb{R})$; extend that argument to every $n \geq 2$, and say which of the three conditions of Proposition 5.6 the set does satisfy.

Exercise 5.21. Show that the set of twice-differentiable functions satisfying $y'' + y = 0$ is a vector space, and verify that sin and cos belong to it. Is the set of solutions of $y'' + y = 1$ also a vector space? Relate your answer to Theorem 3.14.

Exercise 5.22. Give a 2×2 matrix A and a single row operation after which the column space is different. Then explain why no row operation can change $N(A)$ or $R(A)$.

Exercise 5.24. Let $U = \operatorname{span}\{(1,0,0),(0,1,0)\}$ and $W = \operatorname{span}\{(0,1,0),(0,0,1)\}$ in $\mathbb{R}^3$. Describe $U \cap W$ and $U + W$ explicitly, and check that neither $U \cup W$ nor the set difference $U \setminus W$ is a subspace.

Chapter 6: Independence, Basis and Dimension

Exercise 6.23. Decide whether each set is independent, and exhibit a relation when it is not.

(a) $(1,0,1)$, $(0,1,1)$, $(1,1,0)$ in $\mathbb{R}^3$.

(b) $(1,2,3)$, $(4,5,6)$, $(7,8,9)$ in $\mathbb{R}^3$.

(c) $1+t$, $1-t$, t^2 in P_2.

(d) $\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ in $M_2(\mathbb{R})$.

Exercise 6.24. Find a basis and the dimension of each subspace.

(a) $\{(x,y,z) \in \mathbb{R}^3 : x + y + z = 0\}$.

(b) The symmetric 2×2 real matrices.

(c) The polynomials $p \in P_3$ with $p(1) = 0$.

(d) $\operatorname{span}\{(1,1,0,0),(0,1,1,0),(1,2,1,0)\}$ in $\mathbb{R}^4$.

Exercise 6.25. For $A = \left(\begin{smallmatrix}1&2&3&1\\2&4&7&4\\1&2&4&3\end{smallmatrix}\right)$, find bases for $R(A)$, $C(A)$ and $N(A)$, state $\operatorname{rank} A$, and check Corollary 6.17 numerically.

Exercise 6.26. Let $B = \big((2,1),(1,3)\big)$. Verify that B is a basis of $\mathbb{R}^2$ and compute $[x]_B$ for $x = (5,5)$ and for $x = (1,0)$.

Exercise 6.27. Extend $(1,1,0,0)$ and $(0,1,1,0)$ to a basis of $\mathbb{R}^4$, following the procedure in the proof of Theorem 6.14(e). How many different answers are possible?

Exercise 6.28. Let $\dim V = 4$ and let $S \subseteq V$ have four elements. Explain why checking that S is independent is enough to conclude that S is a basis. Then show that the count cannot be dropped: give four independent vectors in $\mathbb{R}^5$ that are not a basis, and five vectors that span $\mathbb{R}^4$ and are not a basis.

Exercise 6.30. Show that 1, $\cos t$, $\cos 2t$ are independent as functions on $\mathbb{R}$, and that 1, $\cos^2 t$, $\cos 2t$ are not. *Hint: for the first, evaluate at three well-chosen points; for the second, recall a double-angle identity.*

Exercise 6.31. Let U and W be subspaces of a finite-dimensional V with $\dim U + \dim W > \dim V$. Show that $U \cap W \neq \{\mathbf{0}\}$. *Hint: take bases of U and of W and count what Theorem 6.14(a) says about their union.*